AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF COMPUTING CHENNAI

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**23CSE304 Embedded Systems Laboratory**

**Activity Sheet #5**

**Activity Name:** DAC Module Interfacing

| | |
|---|---|
| **Name of Student:** | **Chetan Kumar G** |
| **Registration No.:** | **CH.SC.U4CSE24109** |
| **Activity Date:** | **08\|09\|2026** |
| **Submission Date:** | **08\|09\|2026** |

**Assessment Rubrics:**

| Component | Maximum Marks | Marks Allotted |
|---|---|---|
| Program | 3 | |
| Output | 4 | |
| Viva | 2 | |
| Timely Submission | 1 | |
| **Total** | **10** | |

**Faculty In-charge**

**Write an embedded C program for ARM architecture to drive DAC module. Debug and verify the program using IDE on a development computer system.**

## PROGRAM:

```
#include <LPC21xx.H>

#define DACR (*((volatile unsigned long *)0xE006C000)) // DAC register address

void delay(unsigned int t) { unsigned int i;

for(i = 0; i < t; i++);

}

int main(void) { unsigned int value;

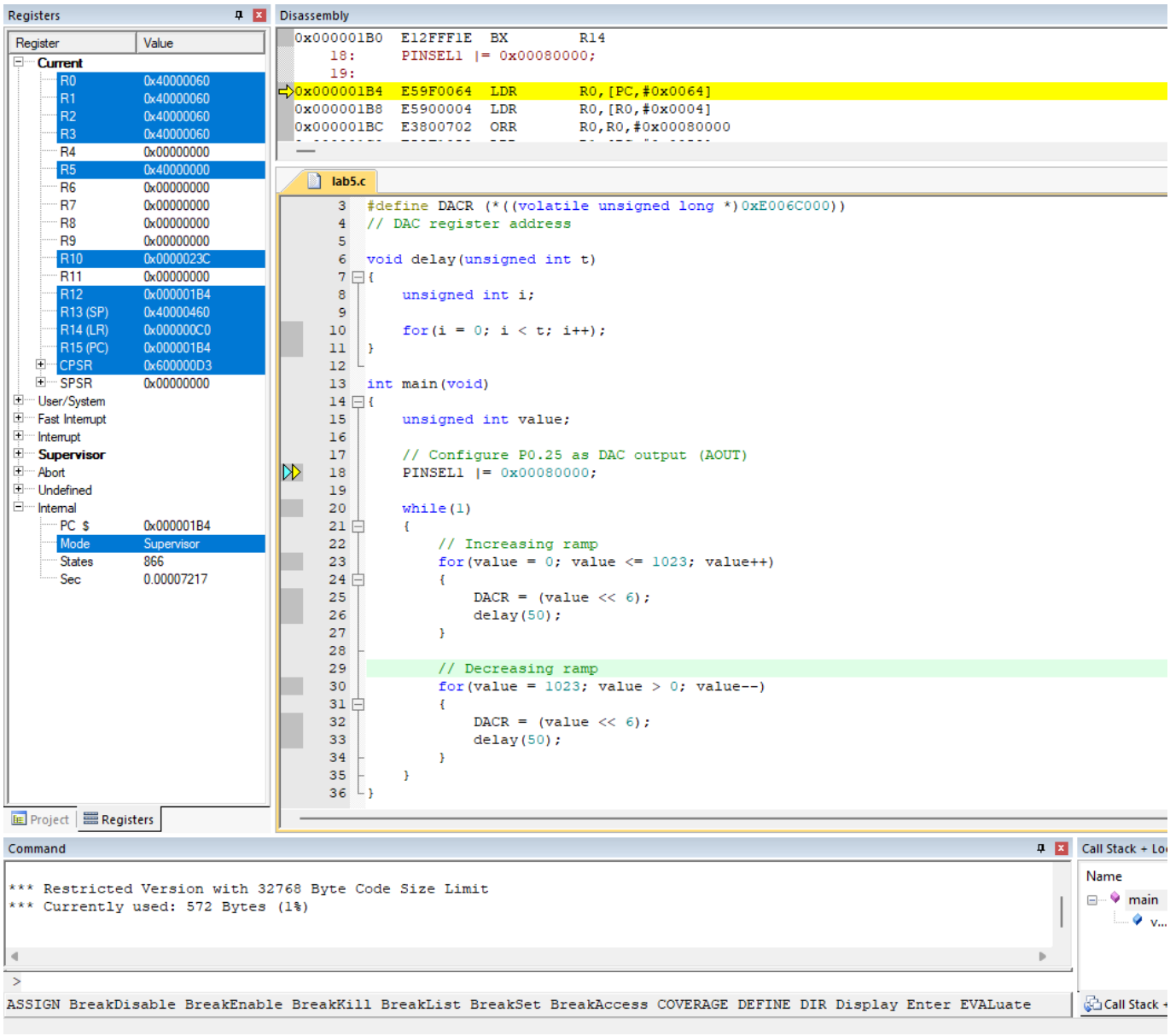
// Configure P0.25 as DAC output (AOUT)
PINSEL1 |= 0x00080000;

while(1)
{
   // Increasing ramp
   for(value = 0; value <= 1023; value++)
   {
      DACR = (value << 6);
      delay(50);
   }

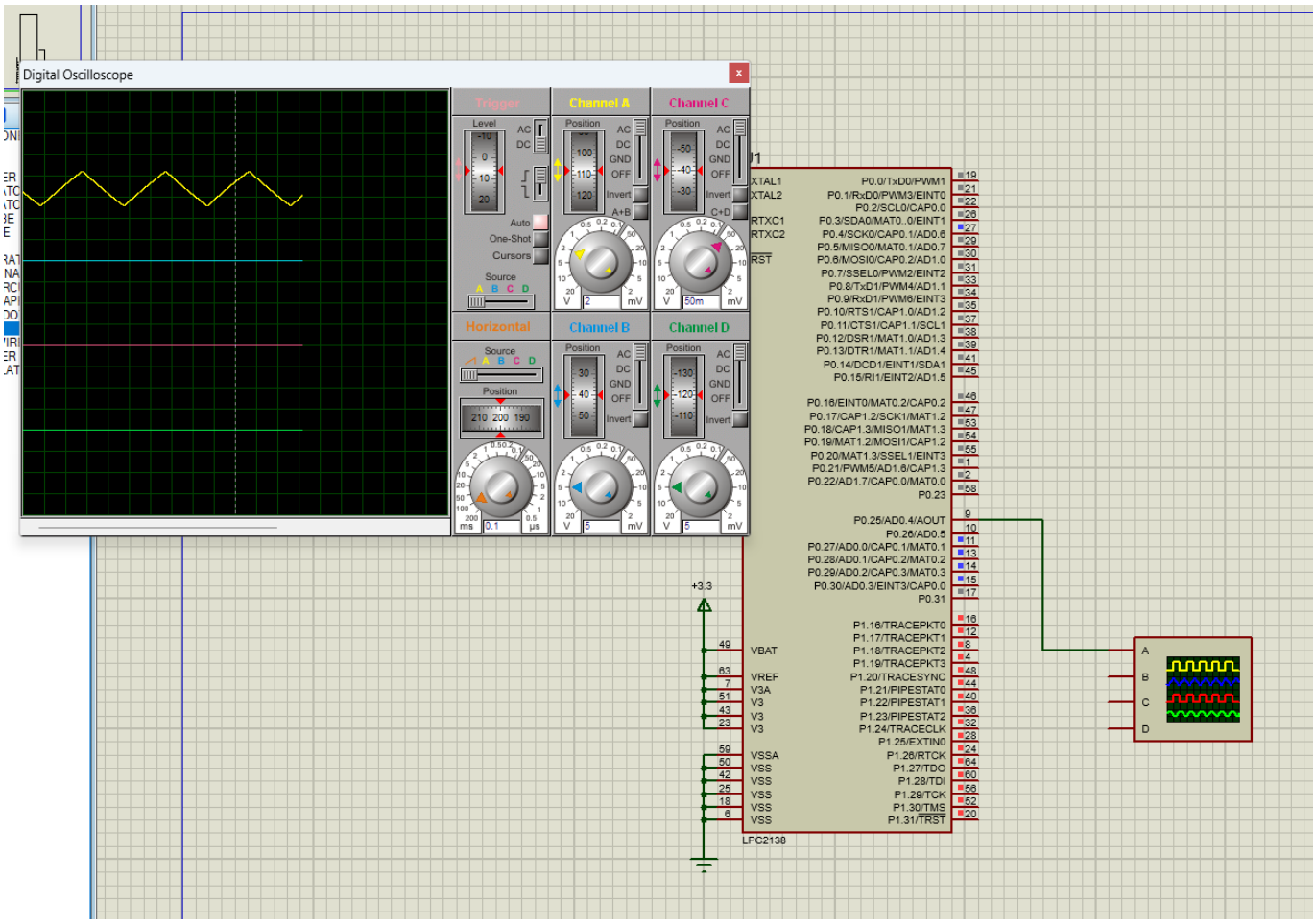
   // Decreasing ramp
   for(value = 1023; value > 0; value--)
   {
      DACR = (value << 6);
      delay(50);
   }
}

}
```

## OUTPUT WINDOW:


Registers
Register
Value
Current
R0 0x40000060
R1 0x40000060
R2 0x40000060
R3 0x40000060
R4 0x00000000
R5 0x40000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x0000023C
R11 0x00000000
R12 0x000001B4
R13 (SP) 0x40000460
R14 (LR) 0x000000C0
R15 (PC) 0x000001B4
CPSR 0x600000D3
SPSR 0x00000000
User/System
Fast Interrupt
Interrupt
Supervisor
Abort
Undefined
Internal
PC $ 0x000001B4
Mode Supervisor
States 866
Sec 0.00007217
Project
Registers
Disassembly
0x000001B0 E12FFF1E BX R14
18: PINSEL1 |= 0x00080000;
19:
0x000001B4 E59F0064 LDR R0,[PC,#0x0064]
0x000001B8 E5900004 LDR R0,[R0,#0x0004]
0x000001BC E3800702 ORR R0,R0,#0x00080000
lab5.c
3 #define DACR (*((volatile unsigned long *)0xE006C000))
4 // DAC register address
5
6 void delay(unsigned int t)
7 {
8 unsigned int i;
9
10 for(i = 0; i < t; i++);
11 }
12
13 int main(void)
14 {
15 unsigned int value;
16
17 // Configure P0.25 as DAC output (AOUT)
18 PINSEL1 |= 0x00080000;
19
20 while(1)
21 {
22 // Increasing ramp
23 for(value = 0; value <= 1023; value++)
24 {
25 DACR = (value << 6);
26 delay(50);
27 }
28
29 // Decreasing ramp
30 for(value = 1023; value > 0; value--)
31 {
32 DACR = (value << 6);
33 delay(50);
34 }
35 }
36 }
Command
*** Restricted Version with 32768 Byte Code Size Limit
*** Currently used: 572 Bytes (1%)
ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE DEFINE DIR Display Enter EVALuate
Call Stack + Lo
Name
main
Call Stack

# OUTPUT:


Digital Oscilloscope
Trigger
Channel A
Channel C
Horizontal
Channel B
Channel D
Level
Position
Source
Auto
One-Shot
Cursors
Invert
LPC2138
XTAL1
XTAL2
RTXC1
RTXC2
RST
VBAT
VREF
V3A
V3
VSSA
VSS
+3.3
A
B
C
D